

MARCUS DIETHELM

Research Engineer

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PROFESSIONAL SUMMARY

Experienced Research Engineer with 7 years of expertise in machine learning systems, data architecture, and advanced algorithm development. Proven track record of translating complex research into production-ready solutions across fintech and cloud infrastructure domains. Skilled at leading cross-functional initiatives and mentoring junior engineers while maintaining hands-on involvement in technical implementation.

WORK EXPERIENCE

Senior Research Engineer

TensorVault Systems

<https://tensorvault.io>

Zurich, Switzerland

March 2022 - Present

Designed and implemented distributed machine learning pipeline processing 500+ GB daily datasets, reducing inference latency by 40% through optimized model compression techniques. Led research initiatives on adversarial robustness for neural networks, resulting in 3 peer-reviewed publications and improved model security protocols. Mentored 4 junior engineers in deep learning best practices and contributed to establishing company-wide ML governance framework. Architected real-time feature engineering infrastructure using Apache Spark and Python, enabling 200+ concurrent model training jobs.

Research Engineer II

CloudInnovate Labs

<https://cloudinnovate.net>

Zurich, Switzerland

June 2019 - February 2022

Developed novel algorithms for distributed graph processing, improving query response times by 35% for enterprise clients handling petabyte-scale datasets. Collaborated with academic partners at ETH Zurich on published research in database optimization, presenting findings at international conferences. Built automated testing suite achieving 94% code coverage for critical ML components, reducing production incidents by 65%. Implemented data pipeline infrastructure supporting real-time analytics for 15+ concurrent stakeholders across organization.

Research Engineer

DataFlow Innovations

<https://dataflow.co>

Zurich, Switzerland

September 2017 - May 2019

Conducted foundational research on time-series anomaly detection algorithms, deploying solutions that identified critical infrastructure failures 48 hours in advance. Developed Python-based prototyping framework reducing

experiment-to-production cycle from 6 months to 3 weeks. Participated in performance optimization initiatives resulting in 50% improvement in data processing throughput. Collaborated with cross-functional teams to document technical specifications and design patterns for internal knowledge base.

EDUCATION

Master of Science in Computer Science

ETH Zurich (Swiss Federal Institute of Technology Zurich)

Zurich, Switzerland

2015 - 2017

Specialization: Machine Learning and Data Systems

Bachelor of Science in Computer Science

University of Zurich

Zurich, Switzerland

2011 - 2015

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, Scala, SQL

Machine Learning: TensorFlow, PyTorch, Scikit-learn, XGBoost, Deep Learning, Neural Networks

Data Processing: Apache Spark, Hadoop, Kafka, Flink, Data Pipelines

Databases: PostgreSQL, MongoDB, Cassandra, Redis, DynamoDB

Cloud Platforms: AWS, Google Cloud Platform, Azure

Software Engineering: Git, Docker, Kubernetes, CI/CD, Linux, Agile

Data Science: Statistical Analysis, Feature Engineering, Model Optimization, A/B Testing

Research: Algorithm Development, Academic Publishing, Technical Writing

LANGUAGES

German (Native)

English (Fluent)

French (Professional Working Proficiency)